

FR1504D Rear Helper Spring Kit Installation Manual

Model: FR1504D (Rear Suspension Package)

Please read these instructions carefully before commencing the installation process. Ensure all standard safety protocols are observed when working beneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Rubber Spacer (4" O.D. x 1" Thick with Bushing)	2
3	Hollow Rubber Spring	2
4	Hex Cap Screw (HCS) (M10 x 40 x 1.5, Grade 8.8, Zinc Plated)	2
5	Hex Cap Screw (HCS) (M10 x 65 x 1.5, Grade 8.8, Zinc Plated)	2
6	Split Lock Washer (M10, Grade 8.8, Zinc Plated)	2
7	Hex Cap Screw (HCS) (M10 x 30 x 1.5, Grade 8.8, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, ensuring the axle is allowed to hang freely. Securely support the chassis before proceeding.
- 2. Remove OEM Bump Stop:** Remove the existing original equipment manufacturer (OEM) bump stop. A 13 mm socket may be required for this operation.
- 3. Assemble Helper Spring Unit:** Assemble the hollow rubber spring (Ref 3) with the rubber spacer (Ref 2) and the weldment bracket (Ref 1) as depicted in the assembly diagram.
- 4. Install Helper Spring Unit:** Mount the complete helper spring assembly to the vehicle frame utilizing the existing mounting holes and the provided fasteners (Ref 4, 5, 6, 7).
- 5. Lower Vehicle:** Carefully lower the vehicle to the ground.

Installation Notes:

- Verify that brake lines or cables do not interfere with the helper spring's position or movement.
- If required, the rubber spacer (Ref 2) and the M10 x 65 hex cap screw (Ref 5) can be utilized to adjust the installed height.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap (Light-Duty):** For light-duty trucks and vans, an unloaded vehicle should exhibit a clearance gap of approximately 1/2 inch to 1-1/2 inches between the helper spring and its contact surface.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or potentially eliminate this initial clearance gap.
- **Heavy-Duty Applications:** For medium and heavy-duty trucks, trailers, and motorhomes, the helper springs are designed to be in constant contact with the axle or frame, resulting in no discernible gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's specified Gross Axle Weight Rating (GAWR).
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under compressive load, the rubber spring's diameter will increase.
- Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.

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